

Exercise 1

Consider a planar 3R robot with unitary link lengths as in Fig. 1 where the generalized coordinates q are defined as the absolute angles of the links w.r.t. the x -axis. The position of the robot end-effector $p = p(q)$, as obtained through the direct kinematics, should follow the desired trajectory

$$p_d(t) = \begin{pmatrix} 1 + 2 \sin 3t \\ 2 + \cos(3t + \frac{\pi}{2}) \end{pmatrix}, \quad \text{for } t \geq 0. \quad (1)$$

The robot is kinematically redundant for this task.

$$p(q) = \begin{cases} c_1 + c_2 + c_3 \\ s_1 + s_2 + s_3 \end{cases} \Rightarrow J = \begin{pmatrix} -s_1 & -s_2 & -s_3 \\ c_1 & c_2 & c_3 \end{pmatrix} \quad p_d(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\dot{p} = J \dot{q} \Rightarrow \ddot{p} = \dot{J} \dot{q} + J \ddot{q} \Rightarrow$$

$$\ddot{p} = \dot{J} \dot{q} + 2 \dot{J} \ddot{q} + J \ddot{q} \quad \dot{J} = \begin{pmatrix} -c_1 \dot{q}_1 & -c_2 \dot{q}_2 & -c_3 \dot{q}_3 \\ -s_1 \dot{q}_1 & -s_2 \dot{q}_2 & -s_3 \dot{q}_3 \end{pmatrix}$$

$$\ddot{J} = \begin{pmatrix} s_1 \dot{q}_1^2 - c_1 \ddot{q}_1 & s_2 \dot{q}_2^2 - c_2 \ddot{q}_2 & s_3 \dot{q}_3^2 - c_3 \ddot{q}_3 \\ -c_1 \dot{q}_1^2 - s_1 \ddot{q}_1 & -c_2 \dot{q}_2^2 - s_2 \ddot{q}_2 & -c_3 \dot{q}_3^2 - s_3 \ddot{q}_3 \end{pmatrix}$$

In a given instant when we have $q, \dot{q}$ and $\ddot{q}$ we evaluate $J, \dot{J}, \ddot{J}$

$$J \ddot{q} = \ddot{p}_d - (\dot{J} \dot{q} + 2 \dot{J} \ddot{q}) \Rightarrow \ddot{q} = J^\# [\ddot{p}_d - (\dot{J} \dot{q} + 2 \dot{J} \ddot{q})]$$

$$\dot{p}_d = \begin{cases} 6 \cos(3t) \\ -3 \sin(3t + \frac{\pi}{2}) \end{cases} \Rightarrow \dot{p}_d(0) = \begin{pmatrix} 6 \\ -3 \end{pmatrix} \quad \begin{array}{c} \text{Diagram of a 3R robot arm with joint angles } q_1, q_2, q_3 \end{array}$$

The initial conf. should be q^* s.t. $p(q) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$q_0 = (0 \quad \frac{\pi}{2} \quad \frac{\pi}{2})^T \quad J_0 = \begin{pmatrix} 0 & -1 & -1 \\ 1 & 0 & 0 \end{pmatrix}$$

$$J_0 \dot{q} = \begin{cases} -\dot{q}_2 - \dot{q}_3 \\ \dot{q}_1 \end{cases} \Rightarrow \dot{q}_0 = (-3 \quad -6 \quad 0)^T \Rightarrow \dot{J}_0 = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \end{pmatrix}$$

$$\ddot{p}_d = \begin{cases} -18 \sin(3t) \\ -9 \cos(3t + \frac{\pi}{2}) \end{cases} \quad \ddot{p}_d(0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \ddot{p}_{d0}$$

$$\ddot{p}_{d0} = \dot{J}_0 \dot{q}_0 + J_0 \ddot{q}_0 \Rightarrow \begin{cases} -9 - \ddot{q}_2 - \ddot{q}_3 = 0 \\ -36 + \ddot{q}_1 = 0 \end{cases} \Rightarrow \ddot{q}_0 = (36 \quad -9 \quad 0)^T$$

$$\ddot{p}_d = \begin{cases} -54 \cos(3t) \\ 18 \sin(3t + \frac{\pi}{2}) \end{cases} \quad \ddot{p}_{d0} = \begin{cases} -54 \\ 18 \end{cases} \quad \ddot{J}_0 = \begin{pmatrix} -36 & 36 & 0 \\ 9 & 9 & 0 \end{pmatrix} \begin{matrix} -3 \\ -6 \\ 0 \end{matrix}$$

$$\ddot{p}_{d0} - (\dot{J}_0 \dot{q}_0 + 2 \dot{J}_0 \ddot{q}_0) = \begin{cases} -54 - 108 + 2(108) = 54 \\ 18 - 81 - 2(54) = -171 \end{cases}$$

$$\Rightarrow \ddot{q}(0) = J^\#(q_0) \begin{pmatrix} 54 \\ -171 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1/2 & 0 \end{pmatrix} = (-171 \quad -27 \quad -27)^T$$

let's assume an error in position, we just have to add a feedback term in the controller:

$$\ddot{\mathbf{q}} = \mathbf{J}^\# [\ddot{\mathbf{p}}_d - (\ddot{\mathbf{J}}\dot{\mathbf{q}} + 2\dot{\mathbf{J}}\ddot{\mathbf{q}})] + \mathbf{K}_P \mathbf{e} + \mathbf{K}_D \dot{\mathbf{e}} + \mathbf{K}_A \ddot{\mathbf{e}} \quad \text{with } K_P, K_D, K_A \succ 0$$

Exercise 2

For the same robot in Fig. 1 and using the same coordinates defined therein, assume that the three links have equal, uniformly distributed mass $m_i = m = 10$ kg, for $i = 1, 2, 3$. Each torque τ_i delivered by the motors and performing work on the absolute coordinate q_i is bounded as $|\tau_i| \leq T_{max} = 300$ Nm, for $i = 1, 2, 3$. Consider the Cartesian regulation control law

$$\tau = \mathbf{J}^T(\mathbf{q}) \mathbf{K}_P (\mathbf{p}_d - \mathbf{p}(\mathbf{q})) - \mathbf{K}_D \dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}), \quad \text{with } \mathbf{p}_d = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad (2)$$

where the gain matrices $\mathbf{K}_P$ and $\mathbf{K}_D$ are diagonal and positive definite. Let the robot starts at rest at $t = 0$ in the configuration $\mathbf{q}(0) = (\pi/2 \ 0 \ 0)^T$.

- If the gain matrices are of the form $\mathbf{K}_P = k_P \cdot \mathbf{I}_{2 \times 2}$ and $\mathbf{K}_D = k_D \cdot \mathbf{I}_{2 \times 2}$, provide the largest values for the scalars k_P and k_D such that $\tau(0)$ in (2) does not violate its bounds.
- Let now the positional gain matrix be $\mathbf{K}_P = \text{diag}\{k_{P_x}, k_{P_y}\}$, while $\mathbf{K}_D$ is as before. Provide the largest values for the scalars k_{P_x} , k_{P_y} , and k_d such that $\tau(0)$ in (2) does not violate its bounds.
- How would things change if the bounds were set as $|\tau_{p,i}| \leq T_{max} = 300$ Nm, where $\tau_{p,i}$ is the torque delivered by the motors and performing work on the relative (Denavit-Hartenberg) coordinate θ_i , for $i = 1, 2, 3$?

The CoM of each link is:

$$\mathbf{p}_{c1} = \begin{cases} 1/2 C_1 \\ 1/2 S_1 \end{cases} \quad \mathbf{p}_{c2} = \begin{cases} C_1 + 1/2 C_2 \\ S_1 + 1/2 S_2 \end{cases} \quad \mathbf{p}_{c3} = \begin{cases} C_1 + C_2 + 1/2 C_3 \\ S_1 + S_2 + 1/2 S_3 \end{cases}$$

$$U = m g_0 \left(\frac{1}{2} S_1 + S_1 + \frac{1}{2} S_2 + S_1 + S_2 + \frac{1}{2} S_3 \right) = m g_0 \left(\frac{5}{2} S_1 + \frac{3}{2} S_2 + \frac{1}{2} S_3 \right)$$

$$\mathbf{g} = \begin{cases} \frac{5}{2} m g_0 C_1 \\ \frac{3}{2} m g_0 C_2 \\ \frac{1}{2} m g_0 C_3 \end{cases} \quad \mathbf{g}(\mathbf{q}(0)) = \begin{pmatrix} 0 & \frac{3}{2} m g_0 & \frac{1}{2} m g_0 \end{pmatrix}^T = \begin{pmatrix} 0 & 147 & 49 \end{pmatrix}^T$$

$$\Rightarrow \mathbf{z} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} k_P & 0 \\ 0 & k_P \end{bmatrix} \left(\begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right) + \begin{pmatrix} 0 \\ 147 \\ 49 \end{pmatrix} \quad \begin{matrix} K_D \text{ is not relevant since} \\ \text{we are at rest.} \end{matrix}$$

$$: \begin{bmatrix} -k_P & 0 \\ 0 & k_P \\ 0 & k_P \end{bmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 147 \\ 49 \end{pmatrix} = \begin{cases} k_P \leq 300 \\ k_P + 147 \leq 300 \\ k_P + 49 \leq 300 \end{cases} \Rightarrow k_P + 147 \leq 300 \Rightarrow k_P \leq 153.$$

$$\text{If } \mathbf{K}_P = \begin{pmatrix} k_{P_x} & 0 \\ 0 & k_{P_y} \end{pmatrix} \text{ we have } \begin{bmatrix} -k_{P_x} & 0 \\ 0 & k_{P_y} \\ 0 & k_{P_y} \end{bmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 147 \\ 49 \end{pmatrix} = \begin{cases} k_{P_x} \leq 300 \\ k_{P_y} + 147 \leq 300 \\ k_{P_y} + 49 \leq 300 \end{cases} \Rightarrow \begin{matrix} k_{P_x} \leq 300 \\ k_{P_y} \leq 153 \end{matrix}$$

We set a coordinate transformation

$$\begin{cases} q_1 = \theta_1 \\ q_2 = \theta_1 + \theta_2 \\ q_3 = \theta_1 + \theta_2 + \theta_3 \end{cases} \Rightarrow \begin{cases} \theta_1 = q_1 \\ \theta_2 = q_2 - q_1 \\ \theta_3 = q_3 - q_2 \end{cases} \quad \mathbf{A} = \frac{\partial \boldsymbol{\theta}}{\partial \mathbf{q}} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

$$\mathbf{g}_\theta = \mathbf{A}^T \mathbf{g} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 147 \\ 49 \end{pmatrix} = \begin{pmatrix} -147 \\ 98 \\ 49 \end{pmatrix}$$

$$\dot{\theta} = A \dot{q} \quad \dot{p} = J \dot{q} = J \bar{A}^{-1} \dot{\theta} \Rightarrow J_{\theta} \dot{\theta}$$

$$\Rightarrow \dot{q} = \bar{A}^{-1} \dot{\theta}$$

$$J_{\theta} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} -1 & -1 & -1 \\ 0 & 1 & 2 \end{pmatrix}$$

$$\Rightarrow \tau = \begin{pmatrix} -1 & 0 \\ -1 & 1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} K_{px} & 0 \\ 0 & K_{py} \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} -147 \\ 98 \\ 49 \end{pmatrix}$$

$$\begin{cases} |K_{px} - 147| \leq 300 \\ |K_{px} + K_{py} + 98| \leq 300 \\ |K_{px} + 2K_{py} + 49| \leq 300 \end{cases} \Rightarrow \begin{cases} K_{px} \leq 153 \\ K_{py} \leq 49 \end{cases}$$

Exercise 3

Consider the planar PRP robot in Fig. 2

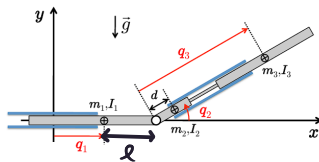


Figure 2: A planar PRP robot moving in a vertical plane, with definition of the generalized coordinates $q = (q_1, q_2, q_3)$ to be used.

- Determine the expressions of the inertial, Coriolis and centrifugal, and gravity terms in the dynamic model expressed in the usual Lagrangian form

$$M(q)\ddot{q} + c(q, \dot{q}) + g(q) = \tau.$$

- Find a factorization $c(q, \dot{q}) = C(q, \dot{q})\dot{q}$ such that $\dot{M} - 2C$ is a skew-symmetric matrix.
- Find all equilibrium configurations q_e (i.e., such that $g(q_e) = 0$), if any.
- Provide symbolic expressions for the scalar coefficients $\alpha > 0$ and $\beta > 0$ such that the following global linear bound holds for the Hessian of the gravitational potential energy $U_g(q)$:

$$\left\| \frac{\partial^2 U_g(q)}{\partial q^2} \right\| = \left\| \frac{\partial g(q)}{\partial q} \right\| \leq \alpha + \beta |q_3|, \quad \forall q \in \mathbb{R}^3.$$

$$p_{c1} = \begin{Bmatrix} q_1 \\ 0 \end{Bmatrix} \Rightarrow T_1 = \frac{1}{2} m_1 \dot{q}_1^2 \quad p_{c2} = \begin{Bmatrix} q_1 + l + d c_2 \\ d s_2 \end{Bmatrix} \quad v_{c2} = \begin{Bmatrix} -d s_2 \dot{q}_2 + \dot{q}_1 \\ d c_2 \dot{q}_2 \end{Bmatrix} \quad \omega_2 = \dot{q}_2$$

$$v_{c2}^T v_{c2} = d^2 \dot{q}_2^2 + \dot{q}_1^2 - 2 d s_2 \dot{q}_1 \dot{q}_2 \Rightarrow T_2 = \frac{1}{2} m_2 (d^2 \dot{q}_2^2 + \dot{q}_1^2 - 2 d s_2 \dot{q}_1 \dot{q}_2) + \frac{1}{2} I_2 \dot{q}_2^2$$

$$p_{c3} = \begin{Bmatrix} q_1 + l + q_3 c_2 \\ q_3 s_2 \end{Bmatrix} \quad v_{c3} = \begin{Bmatrix} \dot{q}_1 + \dot{q}_3 c_2 - q_3 s_2 \dot{q}_2 \\ \dot{q}_3 s_2 + q_3 c_2 \dot{q}_2 \end{Bmatrix}$$

$$v_{c3}^T v_{c3} = \dot{q}_1^2 + \dot{q}_3^2 c_2^2 + q_3^2 s_2^2 \dot{q}_2^2 + 2 c_2 \dot{q}_1 \dot{q}_3 - 2 q_3 s_2 \dot{q}_1 \dot{q}_2 + \dot{q}_3^2 s_2^2 + q_3^2 c_2^2 \dot{q}_2^2$$

$$= \dot{q}_1^2 + \dot{q}_3^2 + q_3^2 \dot{q}_2^2 + 2 c_2 \dot{q}_1 \dot{q}_3 - 2 q_3 s_2 \dot{q}_1 \dot{q}_2$$

$$T_3 = \frac{1}{2} m_3 (\dot{q}_1^2 + \dot{q}_3^2 + q_3^2 \dot{q}_2^2 + 2 c_2 \dot{q}_1 \dot{q}_3 - 2 q_3 s_2 \dot{q}_1 \dot{q}_2) + \frac{1}{2} I_3 \dot{q}_2^2$$

$$M_{11} = m_1 + m_2 + m_3 = c_1$$

$$M_{22} = I_2 + m_2 d^2 + I_3 + m_3 q_3^2 = c_3 q_3^2 + c_4$$

$$M_{12} = -m_2 s_2 - q_3 s_2 = c_2 s_2 - q_3 s_2$$

$$M_{23} = 0$$

$$M_{13} = m_3 c_2 = c_3 c_2$$

$$M_{33} = m_3 = c_3$$

$$M = \begin{bmatrix} c_1 & c_2 s_2 - q_3 s_2 & c_3 c_2 \\ c_2 s_2 - q_3 s_2 & c_3 q_3^2 + c_4 & 0 \\ c_3 c_2 & 0 & c_3 \end{bmatrix}$$

the potential energy is:

$$U = g_0 [m_2 d s_2 + m_3 q_3 s_2] = g_0 s_2 (m_2 d + m_3 q_3)$$

$$g_1 = 0 \quad g_2 = g_0 c_2 (m_2 d + m_3 q_3) \quad g_3 = g_0 s_2 m_3. \quad \text{Now i compute } c_i = \dot{q}^T C_i \dot{q}$$

$$C_1 = \frac{1}{2} \left(\frac{\partial M_1}{\partial \dot{q}} + \left(\frac{\partial M_1}{\partial \dot{q}} \right)^T - \frac{\partial M}{\partial \dot{q}_1} \right) = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 \\ 0 & c_2 (e_3 - q_3) & s_2 (e_3 - 1) \\ 0 & s_2 (e_3 - 1) & 0 \end{pmatrix} \Rightarrow C_1 = \frac{1}{2} c_2 (e_3 - q_3) \dot{q}_2^2 + s_2 (e_3 - 1) \dot{q}_2 \dot{q}_3$$

$$C_2 = \frac{1}{2} \left(\frac{\partial M_2}{\partial \dot{q}} + \left(\frac{\partial M_2}{\partial \dot{q}} \right)^T - \frac{\partial M}{\partial \dot{q}_2} \right) = \begin{pmatrix} 0 & 0 & \frac{1}{2} s_2 (e_3 - 1) \\ 0 & 0 & e_3 q_3 \\ \frac{1}{2} s_2 (e_3 - 1) & e_3 q_3 & 0 \end{pmatrix} \Rightarrow C_2 = 2 e_3 q_3 \dot{q}_2 \dot{q}_3 - 2 (e_3 - 1) \dot{q}_1 \dot{q}_3$$

$$C_3 = \frac{1}{2} \left(\frac{\partial M_3}{\partial \dot{q}} + \left(\frac{\partial M_3}{\partial \dot{q}} \right)^T - \frac{\partial M}{\partial \dot{q}_3} \right) = \begin{pmatrix} 0 & \frac{1}{2} s_2 (1 - e_3) & 0 \\ \frac{1}{2} s_2 (1 - e_3) & -e_3 q_3 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow C_3 = s_2 (1 - e_3) \dot{q}_1 \dot{q}_2 - e_3 q_3 \dot{q}_2^2$$

$$C = \begin{pmatrix} \frac{1}{2} c_2 (e_3 - q_3) \dot{q}_2^2 + s_2 (e_3 - 1) \dot{q}_2 \dot{q}_3 \\ 2 e_3 q_3 \dot{q}_2 \dot{q}_3 - 2 (e_3 - 1) \dot{q}_1 \dot{q}_3 \\ s_2 (1 - e_3) \dot{q}_1 \dot{q}_2 - e_3 q_3 \dot{q}_2^2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & \frac{1}{2} c_2 (e_3 - q_3) \dot{q}_2 & s_2 (e_3 - 1) \dot{q}_2 \\ -2 (e_3 - 1) \dot{q}_1 & 0 & 2 e_3 q_3 \dot{q}_2 \\ s_2 (1 - e_3) \dot{q}_2 & -e_3 q_3 \dot{q}_2 & 0 \end{pmatrix}}_C \dot{q}$$

$$\begin{cases} g_2 = g_0 c_2 (e_3 + e_3 q_3) = 0 \\ g_3 = e_3 g_0 s_2 = 0 \end{cases} \Rightarrow \begin{cases} q_3 = \pm g_0 \frac{e_3}{e_3} \\ q_2 \in \{0, \pi\} \end{cases}$$

$$\text{let } q' = \begin{pmatrix} q_2 \\ q_3 \end{pmatrix} \quad A = \frac{\partial g_i}{\partial q'} \quad q' = (q_2 \ q_3)^T$$

$$A = \begin{pmatrix} -g_0 s_2 (e_3 + e_3 q_3) & g_0 c_2 e_3 \\ e_3 g_0 c_2 & 0 \end{pmatrix}$$

$$\lambda I - A^T A = \begin{pmatrix} \lambda + g_0^2 s_2^2 (e_3 + e_3 q_3)^2 + e_3^2 g_0^2 c_2^2 & g_0^2 c_2 s_2 e_3 (e_3 + e_3 q_3) \\ g_0^2 c_2 s_2 e_3 (e_3 + e_3 q_3) & \lambda - g_0^2 c_2^2 e_3^2 \end{pmatrix}$$

$$\det(\lambda I - A^T A) = (\lambda + \alpha)(\lambda - \delta) - \beta^2$$

$$= \lambda^2 + \lambda(\alpha - \delta) - \delta\alpha - \beta^2$$

$$\Delta = (\alpha - \delta)^2 - 4\delta\alpha - \beta^2 \Rightarrow \lambda_{1,2} = \frac{\delta - \alpha \pm \sqrt{\Delta}}{2} \Rightarrow \sqrt{\lambda_{max}} = \left(\frac{\delta - \alpha + \sqrt{\Delta}}{2} \right)^{1/2} \Rightarrow$$

upper bound of $\|A\|$